Recall

Work from memory, then check with your tutor.

R1. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

Answer: _____

R2. Find $10 - 2.286$.

Answer: _____

R3. Solve $x + 9 = 23$.

Answer: _____

R4. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

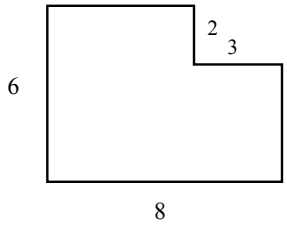
S. Return here near the end of the lesson

To find surface area, I list __ and add __.

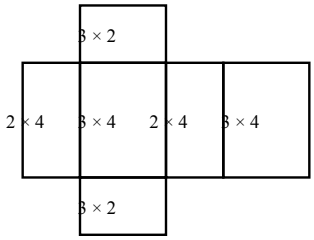
Nets and surface area

Follow the slide labels. P = problem; F = your turn; D = discussion.

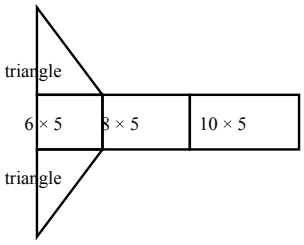
P1. An 8 cm by 6 cm rectangle loses a 3 cm by 2 cm corner from the upper right. Find the L-shaped area two ways.



P2. A closed box is 2 cm by 3 cm by 4 cm. Label all six rectangles on a net. Find surface area and volume.



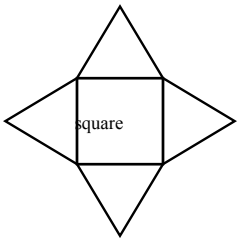
P3. A triangular prism has right-triangle ends with sides 6, 8, and 10 cm. Its length is 5 cm. List the five faces and find surface area.



Nets and surface area

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. A square pyramid has base side 6 cm. Each triangular face has perpendicular height 5 cm. Find its surface area.



P5. A closed 2 cm by 4 cm by 5 cm box has a claimed surface area of 68 square cm. Its 2 by 4 bottom was left out. Repair the answer.

Answer: _____

F. A square pyramid has base side 4 cm and face height 3 cm. Complete: SA = $16 + 4 \times (\frac{1}{2} \times 4 \times __) =$ __.

Answer: _____

Nets and surface area

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Ava finds a cube’s surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

Answer: _____

Mixed review

M1. A ribbon is $4\ \frac{1}{2}$ meters long. Pieces measure $\frac{3}{8}$ meter. How many pieces fit?

Answer: _____

M2. Find 312 ÷ 24.

Answer: _____

M3. Use y = 2x for x = 1, 3, 5. Give the ordered pairs.

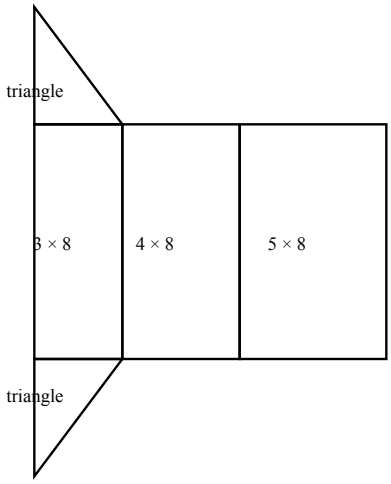
Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A triangular prism has right-triangle ends with sides 3, 4, and 5 cm. It is 8 cm long. Find the area of all five faces together.

Answer: _____



Area and surface area

The big idea

Area counts square units that cover a surface. Cutting and rearranging reveal triangle and parallelogram formulas. A net helps count every face of a solid.

Words to know

area: The amount of flat surface inside a shape. Example: A 2 by 3 rectangle covers 6 square units.

base: The side chosen to match a height. Example: The bottom side can be the base.

height: The perpendicular distance from the base line to the opposite side or vertex. Example: A height meets the base line at a right angle.

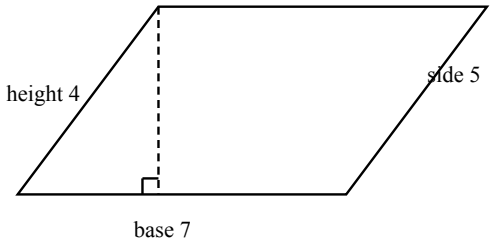
composite polygon: A shape made by joining simpler polygons. Example: An L can split into two rectangles.

net: A flat arrangement that folds into a solid. Example: Six rectangles can form a box.

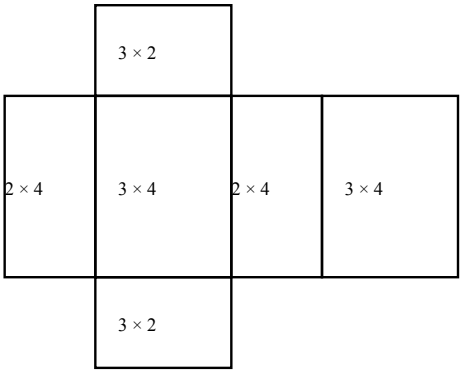
surface area: The total area of all outside faces. Example: A closed box includes its bottom.

volume: The space inside a solid. Example: Volume uses cubic units.

How it works



Cut and slide the left end to make a rectangle. A parallelogram rearranges into a rectangle. Two copies of a triangle make a parallelogram.



Match each rectangle to one face. A net unfolds every face. Surface area adds face areas in square units.

Worked examples

1. A parallelogram has base 6 cm and height 3 cm.

Cut a triangle from one end and move it to the other.

The new rectangle has the same area, base 6 and height 3.

Area = 6 × 3 = 18 square cm.

Answer: 18 square cm

2. Find the surface area of a 2 cm by 2 cm by 3 cm closed box.

Two square ends each have area 4.

Four side rectangles each have area 6.

Total area = 2 × 4 + 4 × 6 = 32 square cm.

Answer: 32 square cm

Common mistakes

Uses a slanted side as height. Why: Treats any given length as interchangeable. Fix: Trace the perpendicular distance with a ruler.

Omits the half for a triangle. Why: Uses the containing parallelogram. Fix: Pair congruent copies and split the area equally.

Uses cm for area. Why: Confuses boundary length and surface coverage. Fix: Count a row of unit squares.

Try it

1. A rectangle is 7 cm by 3 cm. Find its area.

2. A rectangle is 8 cm by 2 cm. Find its area.

3. Find the surface area of a closed 2 by 3 by 5 cm box.

4. Find the surface area of a closed 3 by 3 by 4 cm box.

5. Find the surface area of a closed 2 by 4 by 6 cm box.

Answers

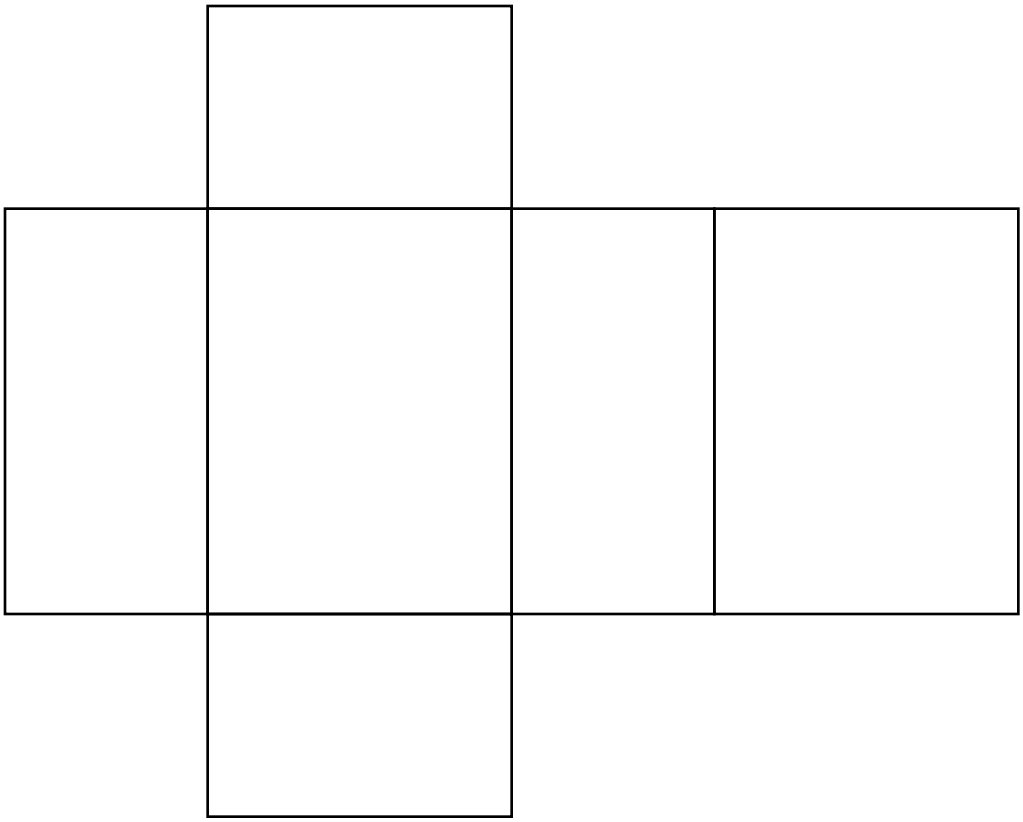
1. 21 square cm 2. 16 square cm 3. 62 square cm 4. 66 square cm 5. 88 square cm

Where this connects

Multiplication and fraction multiplication measure flat regions. Nets connect those regions to three-dimensional solids.

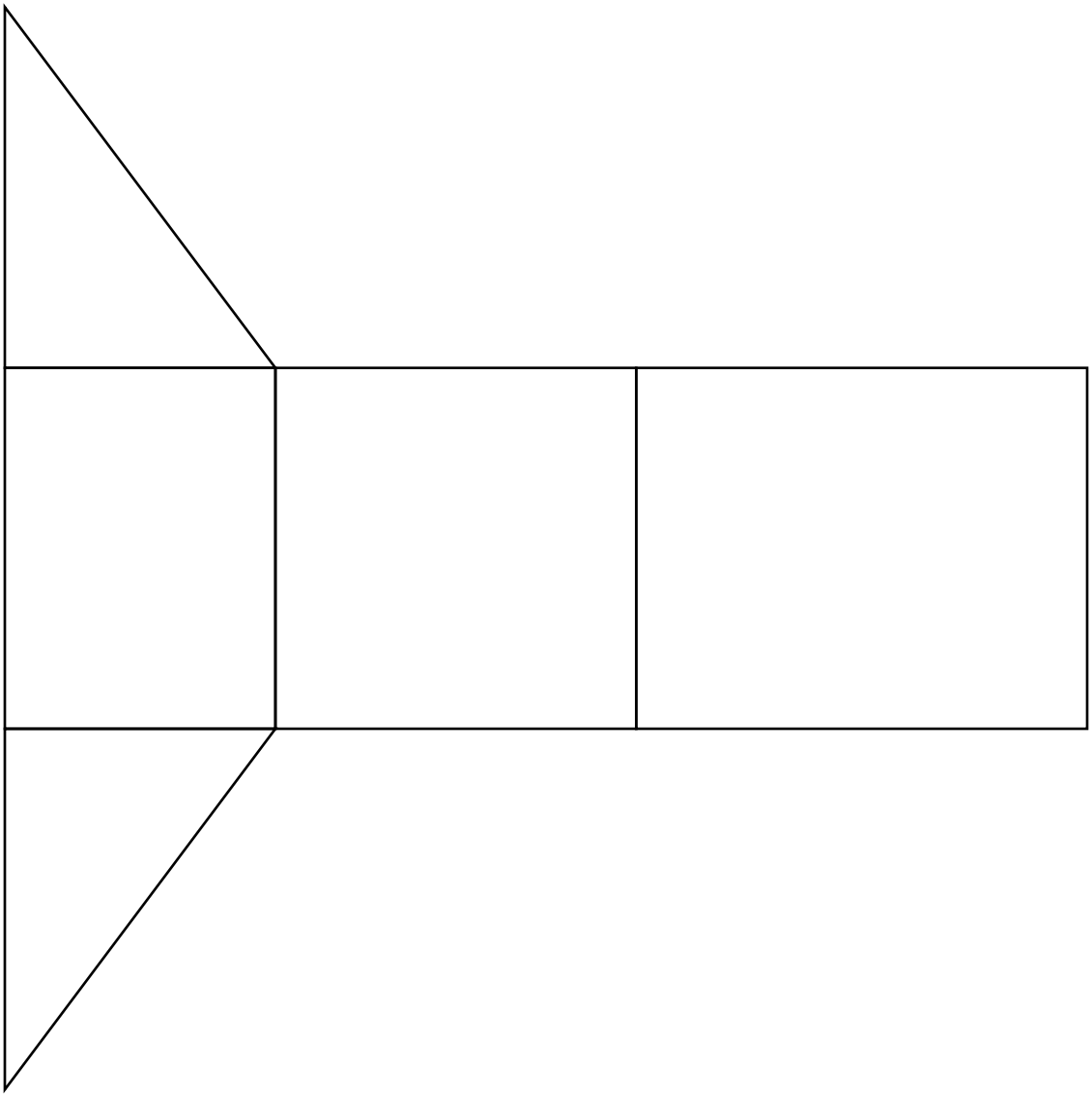
Rectangular prism net

Cut the outside boundary. Fold the internal edges. Tape the faces.



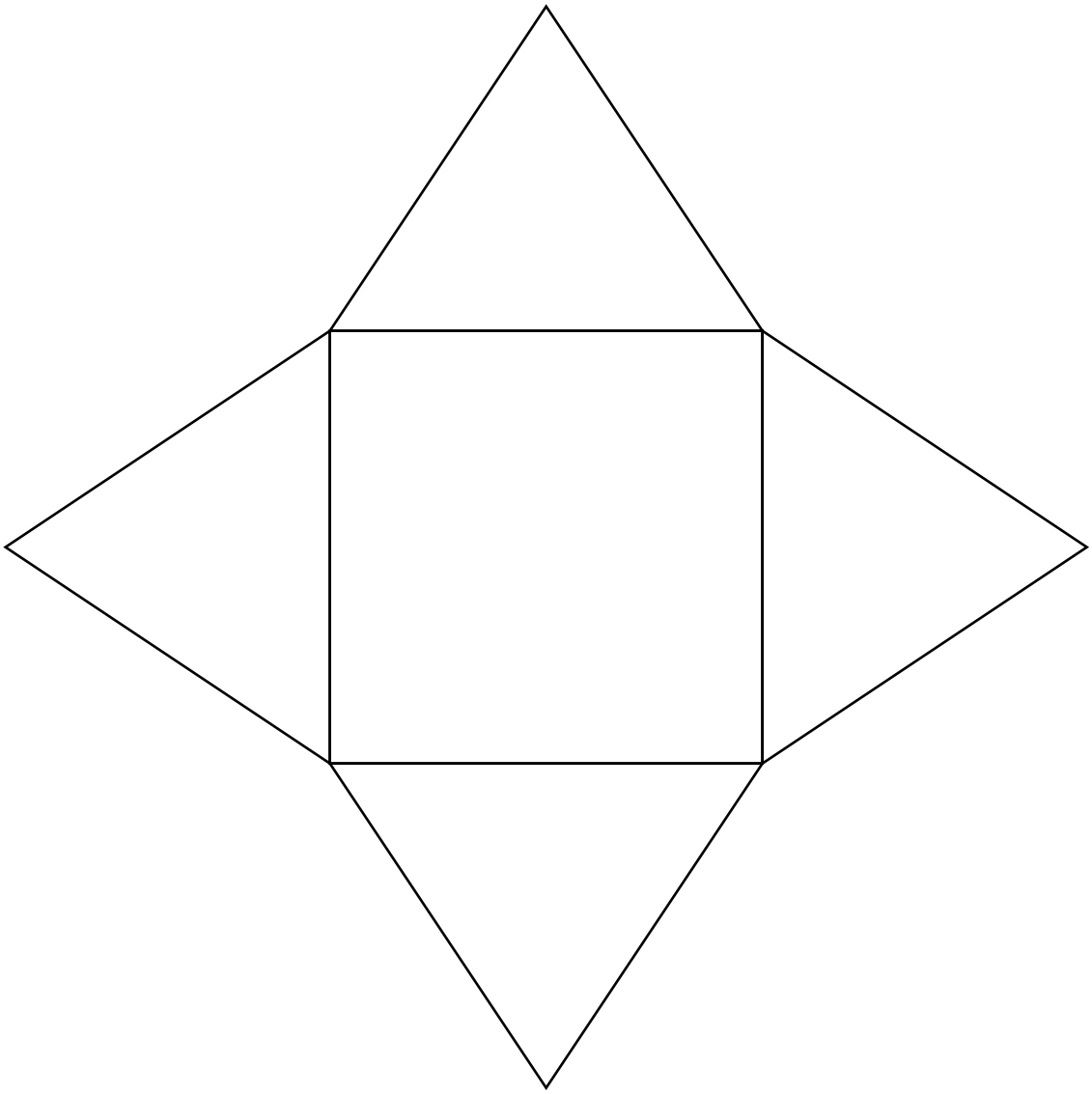
Edges are 2, 3 and 4 units. Shared edges have matching lengths.

Triangular prism net



Triangles: side lengths 3, 4, 5. Prism length: 4 units.

Square pyramid net



Square side: 4 units. Each triangular face has height 3 units.